

Recommendation Mathematics & Decision Science.

The mathematical foundation beneath every recommendation Auxion produces. AIS-001 established that recommendations are *derived, not written*; this document specifies exactly how. It defines the decision models, the scoring functions, and the ranking mathematics that turn validated evidence into a priced, prioritized, defensible set of moves — and proves, with formulas, why one recommendation outranks another and how that ranking shifts as evidence changes.

MCDA + BAYESIAN

RISK-ADJUSTED EV

13-FIELD RECORD

WORKED PROOFS

PORTFOLIO OPTIMIZATION

NOTATION

All scores are normalized to $[0,1]$ unless noted. **C** confidence · **I** impact · **p** probability of success · **E** effort · **R** risk · **U** urgency · **S** strategic alignment · **EV** expected value · π priority. Weights w_k sum to 1.

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01 DECISION THEORY & MCDA

FOUNDATION

WHY: a business decision is a choice among alternatives under uncertainty with multiple, conflicting criteria. Classical single-metric ranking (e.g. ROI alone) is wrong because it ignores risk, confidence, and strategic fit. Auxion scores every recommendation as a multi-criteria decision under uncertainty — the expected value of a normalized, weighted criteria vector.

MASTER SCORE — WEIGHTED MULTI-CRITERIA UTILITY

$$V(\mathbf{x}) = C(\mathbf{x}) \cdot \sum_k w_k \cdot x_k(\mathbf{x})$$

Each criterion x_k is normalized to [0,1]; weights w_k encode the firm's decision policy and sum to 1. The whole weighted sum is scaled by confidence C , so an uncertain recommendation cannot outrank a well-evidenced one on raw appeal alone.

Alternatives

Each recommendation is an alternative in the decision space; the do-nothing baseline is always included as a reference alternative.

Criteria

The scored dimensions — impact, opportunity, risk-reduction, strategic fit — each normalized so they are comparable.

Weights

w_k encode the firm's (or engagement's) policy on what matters; they are explicit, versioned, and auditable.

Normalization

Every criterion is mapped to [0,1] against its observed range so no single unit dominates by scale.

02 BAYESIAN REASONING & CONFIDENCE PROPAGATION

$P(H|E)$

WHY: evidence should update belief, not replace it. Every finding's confidence is a posterior belief that new evidence revises. Confidence then propagates: a conclusion is never more certain than the weakest link it depends on.

BAYESIAN UPDATE

$$P(H|E) = P(E|H) \cdot P(H) / P(E)$$

Prior belief in a hypothesis is updated by the likelihood of the observed evidence. Each new source shifts the posterior; the engine keeps the running posterior as the finding's confidence.

CONFIDENCE PROPAGATION (SERIAL)

$$C(\text{chain}) = \prod_i C_i$$

Confidence through a dependent chain multiplies, so it can only fall. A single weak step ($C \ll 1$) caps the whole conclusion — weakness never launders into certainty.

CORROBORATION (PARALLEL, INDEPENDENT SOURCES)

$$C(\text{joint}) = 1 - \prod_i (1 - C_i)$$

Independent agreeing sources raise confidence: two sources at 0.7 corroborate to 0.91. This is why the swarm seeks a second independent agent before committing a high-stakes finding.

03 EVIDENCE, MISSINGNESS & CONTRADICTION

C AS A FUNCTION

COMPOSITE CONFIDENCE (FROM AIS-001, FORMALIZED)

$$C = cov \cdot rel \cdot fresh \cdot agree \cdot comp$$

coverage · reliability · freshness · agreement · completeness, each in [0,1]. The product form means any factor near zero caps the whole — the engine reports low confidence rather than papering over a gap.

Missing evidence

$$C \leftarrow C \cdot cov$$

Lower coverage multiplies confidence down. A recommendation resting on thin evidence carries visibly lower C and cannot rise to the top on impact alone.

Contradictory evidence

$$agree = 1 - disp$$

Disagreement among sources lowers the agreement factor, cutting confidence and, through π , priority. Contradiction is surfaced, never averaged away.

Stale evidence

$$fresh = e^{(-\lambda \Delta t)}$$

Freshness decays with age at a per-dimension rate, so an old fact contributes less to confidence until refreshed.

04 IMPACT, EXPECTED VALUE & ROI

RISK-ADJUSTED

WHY: a raw impact figure is a fantasy until it is discounted by the probability of achieving it and the confidence in the estimate. Auxion always ranks on risk-adjusted expected value, never on headline impact.

EXPECTED VALUE

$$EV = p \cdot I - (1 - p) \cdot L$$

Probability of success times impact, less the expected loss of failure. A high-impact move with low p can have negative EV.

RISK-ADJUSTED ROI

$$ROI = (EV \cdot C - Cost) / Cost$$

Expected value is further scaled by confidence before net return is taken over cost, yielding a defensible ROI band, not a point.

Business impact

Modeled movement in the affected Business Health Index dimensions, weighted by each dimension's Index weight.

Financial impact

Projected revenue, cost, or margin effect in currency, carried with its confidence band, not as a single number.

ROI prediction

Risk-adjusted expected value net of estimated cost, reported as a range whose width reflects confidence.

05 RISK & OPPORTUNITY MODELING

EXPOSURE

RISK EXPOSURE

$$R = P(\text{event}) \cdot \text{Severity} \cdot (1 - \text{Mitigation})$$

Likelihood times severity, discounted by existing mitigation. Risk propagates along capability edges in the Business Graph.

OPPORTUNITY SIZE

$$O = \text{Gap} \cdot \text{Reachable} \cdot \text{Value} \cdot p$$

The measured gap, the reachable fraction of it, its unit value, and the probability of capture. An unreachable gap sizes to zero.

Severity classes

Exposure is bucketed into ordered severity classes; a critical class triggers the priority override in §06.

Mitigation credit

Existing controls reduce residual risk; the model scores residual exposure, not gross.

Propagation

Risk and opportunity flow along Business Graph edges, so a single-node finding raises exposure on the nodes it endangers or unlocks.

Reachability

Opportunity is discounted by the reachable fraction of the gap; an unreachable gap contributes nothing.

06 PRIORITY, URGENCY & STRATEGIC ALIGNMENT

π

PRIORITY SCORE

$$\pi(\mathbf{x}) = C \cdot (w_I \cdot I + w_O \cdot O + w_R \cdot R + w_S \cdot S) \cdot U / (E + \epsilon)$$

Confidence-scaled, weighted value (impact, opportunity, risk-reduction, strategic fit), amplified by urgency and divided by effort. This single scalar is what the ranking sorts on. Critical risks receive an override that floors their priority above all optimization regardless of effort.

Urgency

$$U = f(\text{decay}, \text{deadline})$$

How fast the value erodes if action is delayed. A closing window or an accelerating risk raises U.

Strategic alignment

$$S = \text{fit to goals}$$

How well the move serves the client's stated objectives; misaligned high-ROI moves are down-weighted.

Effort

$$E = \text{normalized cost/time}$$

Implementation difficulty in the denominator, so cheap wins rise and heavy lifts must justify themselves on value.

07 THE RECOMMENDATION RECORD

13 FIELDS

Every recommendation is a structured record of thirteen fields. Each is either measured from evidence or computed by the models above — none is free-text opinion.

FIELD	TYPE	DERIVATION
Evidence	<code>ref</code>	Links to the exact evidence records supporting the move; inherited from the AIS-001 evidence system.
Confidence	<code>[0,1]</code>	Composite C from §03; gates whether the move may rank at all.
Business impact	<code>[0,1]</code>	Weighted Index-dimension movement if executed.
Financial impact	<code>currency</code>	Projected monetary effect with a confidence band.
Implementation effort	<code>[0,1]</code>	Normalized cost and time to execute.
Dependencies	<code>DAG</code>	Prerequisite moves that must precede this one.
Time horizon	<code>enum</code>	Days · weeks · quarter · quarter+, from effort and dependencies.
Probability of success	<code>[0,1]</code>	p, estimated from evidence and comparable outcomes.
Risk	<code>[0,1]</code>	Residual exposure of acting and of not acting.
Owner	<code>role</code>	The accountable role for execution, assigned at proposal time.
Expected ROI	<code>ratio+band</code>	Risk-adjusted ROI from §04, as a range.
Success metrics	<code>metric set</code>	The measurable signals that confirm the move worked.
Review cycle	<code>cadence</code>	When the move and its assumptions are re-checked against fresh evidence.

08 WHY RECOMMENDATION A OUTRANKS B

WORKED EXAMPLE

Two candidate moves. B has the larger headline impact, yet A ranks higher once confidence, probability, and effort are applied. The math is fully shown.

MOVE	C	I	p	E	O	S
A • Fix checkout friction	0.85	0.70	0.80	0.30	0.60	0.70
B • Rebrand identity	0.55	0.90	0.45	0.80	0.50	0.60

MOVE A — PRIORITY

```
value = .35(.70)+.25(.60)+.20(.70)+.20(.70)
= 0.68
 $\pi = 0.85 \cdot 0.68 \cdot U(1.0) / (0.30+0.05)$ 
 $\pi(A) \approx 0.66$ 
```

MOVE B — PRIORITY

```
value = .35(.90)+.25(.50)+.20(.40)+.20(.60)
= 0.64
 $\pi = 0.55 \cdot 0.64 \cdot U(0.8) / (0.80+0.05)$ 
 $\pi(B) \approx 0.30$ 
```

Result: A ranks above B. Although B's raw impact is higher (0.90 vs 0.70), A's stronger evidence (C 0.85 vs 0.55), higher probability of success, and lower effort give it a priority of **0.66** against B's **0.30**. Auxion never lets an alluring but weakly-evidenced, hard-to-execute move outrank a confident, achievable one.

09 DEPENDENCIES, TRADEOFFS & SENSITIVITY

DAG

WHY: recommendations are not independent. One unlocks another; two compete for the same budget; a small change in a weight can reorder the whole list. The engine models these relationships explicitly.

Dependency graph

Moves form a DAG; a dependent cannot be scheduled before its prerequisites, and its horizon inherits theirs.

Tradeoff analysis

When moves compete for the same budget or resource, the engine surfaces the tradeoff on the efficient frontier rather than hiding one.

Sensitivity analysis

Each weight is perturbed by $\pm\delta$ to test whether the ranking holds; fragile orderings are flagged as close calls.

SENSITIVITY — RANK STABILITY UNDER WEIGHT PERTURBATION

$\text{stable}(r) \Leftrightarrow \arg\max \pi \text{ unchanged for all } w_k \pm \delta$

A ranking is reported as robust only if its order survives small perturbations of every weight. Fragile rankings — those that flip under a small δ — are flagged so an operator knows the call is close.

10 SCENARIOS, COUNTERFACTUALS & FORECAST CONFIDENCE

UNDER UNCERTAINTY

Scenario planning

Compare multiple weight or assumption sets (e.g. growth-first vs risk-first) and show how the ranking changes under each.

Counterfactual reasoning

Estimate the outcome of not acting, or of acting differently, as the baseline every recommendation is measured against.

Forecast confidence

Every forward projection carries a confidence that widens with horizon; long-range forecasts are reported as ranges, never points.

Alternative hypotheses

Where evidence admits more than one explanation, competing hypotheses are scored and the dominant one reported with its margin.

WHEN MONTE CARLO IS WARRANTED

Monte Carlo simulation is applied only when a recommendation's inputs are genuinely distributional and interacting — multiple uncertain variables whose combined effect cannot be found analytically. For a single-variable estimate it is overkill; the engine uses a closed-form confidence band instead. Suitability is a decision the engine makes explicitly, so compute is spent where uncertainty actually compounds.

11 PORTFOLIO OPTIMIZATION

CONSTRAINED SELECTION

WHY: a client cannot execute everything at once. The engine selects the subset of moves that maximizes portfolio value under a budget and effort constraint — a constrained knapsack, respecting dependencies.

SELECTION OBJECTIVE

$$\max \sum \pi(r_i) \cdot s_i \text{ s.t. } \sum \text{Cost}(r_i) \cdot s_i \leq B, s_i \in \{0,1\}$$

Choose the set that maximizes total priority within budget B, with dependency constraints forcing prerequisites in before dependents. Critical risks are pinned in regardless of budget pressure.

Quick Wins

HIGH π / LOW E

Low-effort, high-priority moves selected first for momentum.

Long-Term Initiatives

HIGH I / HIGH E

Structural moves that change trajectory over a quarter or more.

Critical Risks

$R \geq \text{crit}$

Pinned into the portfolio regardless of budget or effort.

Opportunity Sizing

O-RANKED

Growth moves ordered by risk-adjusted opportunity size.

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HOW RECOMMENDATIONS EVOLVE OVER TIME

TEMPORAL

WHY: a recommendation is a claim about a moment. As evidence ages, as moves are executed, and as the market shifts, priorities recompute. The engine treats the recommendation set as a living, versioned object.

FRESHNESS DECAY OF EVIDENCE-BACKED CONFIDENCE

$$C(t) = C_0 \cdot e^{-\lambda(t - t_0)}$$

Confidence decays exponentially with the age of its evidence, at a per-dimension rate λ . A recommendation whose confidence decays below threshold is automatically re-queued for a partial re-scan.

Recompute on re-scan

Each monitoring cycle refreshes evidence, updates C , and re-runs π — the ranking is never frozen.

Execution feedback

As moves are completed, their impact is measured and fed back, adjusting p estimates for similar future moves.

Market drift

Competitor and benchmark shifts change gaps and opportunity sizes, re-ordering the set even with no client change.

Versioned history

Every recommendation-set version is retained, so the trajectory of advice is auditable over the engagement.

A

FORMULA REFERENCE

CANONICAL SET

Master utility	$V = C \cdot \sum w_k \cdot x_k$
Bayesian update	$P(H E) = P(E H) \cdot P(H) / P(E)$
Serial confidence	$C = \prod C_i$
Corroboration	$C = 1 - \prod(1 - C_i)$
Composite confidence	$C = cov \cdot rel \cdot fresh \cdot agree \cdot comp$
Expected value	$EV = p \cdot I - (1-p) \cdot L$
Risk-adjusted ROI	$ROI = (EV \cdot C - Cost) / Cost$
Risk exposure	$R = P \cdot Severity \cdot (1 - Mitigation)$
Opportunity size	$O = Gap \cdot Reachable \cdot Value \cdot p$
Priority	$\pi = C \cdot (\sum w_k \cdot x_k) \cdot U / (E + \epsilon)$
Portfolio	$\max \sum \pi \cdot s_i \text{ s.t. } \sum Cost \cdot s_i \leq B$
Confidence decay	$C(t) = C_0 \cdot e^{(-\lambda \Delta t)}$